

DATA SCIENCE

Complete Learning Roadmap

Transform raw datasets into reliable insights through statistical rigor and experimentation.

Python and notebooks

Hypothesis testing

Distributions

Correlation analysis

Data storytelling

Pandas

01

Fundamentals

02

Core Concepts

03

Tools and Frameworks

04

Real Projects

05

Advanced Topics

5

Learning Stages

20+

Core Topics

10+

Projects

100%

Data

Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Python and notebooks

02

Core Concepts

Hypothesis testing •
Distributions • Correlation
analysis • Data storytelling

03

Tools and Frameworks

Pandas • NumPy •
Seaborn/Plotly • Notebook
pipelines

04

Real Projects

Business KPI dashboard •
Experiment readout project •
Data quality checks • Segment
analysis

05

Advanced Topics

Causal inference basics •
Productionized analytics •
Experiment design
governance • Data ethics

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Python and notebooks
- Data wrangling
- Exploratory analysis
- SQL querying

Tools & Frameworks

- Python and notebooks

Recommended Projects

- Data Science workshop using Python and notebooks
- Data Science portfolio project with Data wrangling

Stage 1 Milestone Complete Data Science workshop using Python and notebooks and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Hypothesis testing
- Distributions
- Correlation analysis
- Data storytelling

Tools & Frameworks

- Hypothesis testing
- Distributions
- Correlation analysis
- Data storytelling

Recommended Projects

- Data Science workshop using Hypothesis testing
- Data Science portfolio project with Distributions

Stage 2 Milestone Complete Data Science workshop using Hypothesis testing and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- Pandas
- NumPy
- Seaborn/Plotly
- Notebook pipelines

Tools & Frameworks

- Pandas
- NumPy
- Seaborn/Plotly
- Notebook pipelines

Recommended Projects

- Data Science workshop using Pandas
- Data Science portfolio project with NumPy

Stage 3 Milestone Complete Data Science workshop using Pandas and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Business KPI dashboard
- Experiment readout project
- Data quality checks
- Segment analysis

Tools & Frameworks

- Business KPI dashboard
- Experiment readout project
- Data quality checks
- Segment analysis

Recommended Projects

- Data Science workshop using Business KPI dashboard
- Data Science portfolio project with Experiment readout project

Stage 4 Milestone Complete Data Science workshop using Business KPI dashboard and document the decisions, trade-offs, and validation checks before moving to Stage 5.



Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

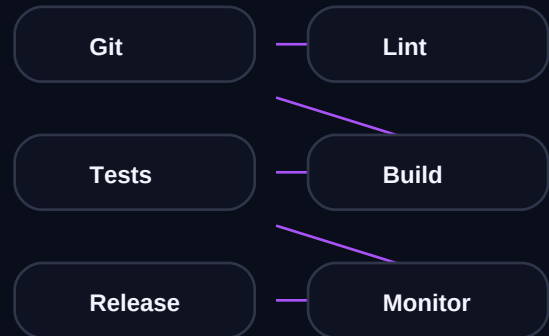
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Causal inference basics
- Productionized analytics
- Experiment design governance
- Data ethics

CI/CD Pipeline



Stage 5 Milestone

A production-grade data science workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Data Science docs and community guides

Link: <https://pandas.pydata.org/docs/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/ai-data-scientist>

PAID

Structured specialization track focused on Data Science

Link: <https://roadmap.sh/ai-data-scientist>

FREE

Data Science roadmap reference

Link: <https://roadmap.sh/ai-data-scientist>

FREE

Official documentation

Link: <https://pandas.pydata.org/docs/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Data Science?
- Which project best proves Stage 4 competency in Data Science?
- What trade-offs appear in Stage 5 decisions for Data Science?